

Loan Fraud detection project Report

CERTIFICATE IN ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING SKILLS

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Introduction

Loan Fraud is a significant concern for financial institutions, leading to substantial financial losses and reputation damage. As fraudsters become more sophisticated, there is a pressing need for a reliable and efficient system to detect and prevent fraudulent activities in loan applications. The Euler Group, comprising Luke Masaba K., Stella Kwera M., David T. Mutebi and Maria Charity Atuhairu, for their capstone project under Refactory, undertook a project to address this challenge. The project aimed at developing a predictive model to accurately determine the likelihood of loan repayment, enhance the efficiency of loan processing by automating fraud detection, reduce financial losses, and strengthen trust between lenders and borrowers.

0.1 problem statement

Super lender, a local digital lending company, leverages credit models to deliver profitable, high-impact loan alternatives. The project focuses on two key drivers of loan default: willingness to pay and ability to pay. This project proposes the development of a Loan Fraud Detection system that leverages artificial intelligence to identify suspicious patterns and improve decision-making in loan processing.

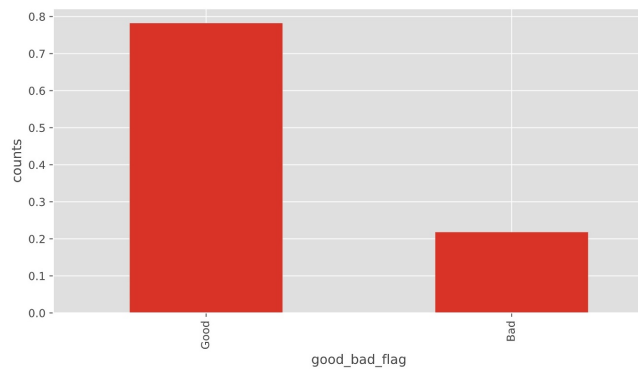
0.2 Objectives

1. Predictive Model Development: To develop a predictive model that accurately determines each application's loan repayment likelihood, whether the loan is bad or good .
2. Automated Fraud Detection: To enhance the efficiency of loan processing by automating the detection of fraudulent activities
3. Risk Reduction: To reduce financial losses and reputational risks for the financial institution, Superlender.
4. Trust Enhancement: To strengthen trust between lenders and customers by ensuring fair and secure lending practices.

0.3 Data Description

The dataset used for this project comprised approximately 14,743 rows and 28 columns, including some columns with null values and duplicates. The data included:

1. Demographics Data: Personal information about borrowers, such as age, employment status, location, and education level.
2. Previous Loans Data: Historical information on previously accessed loans, including loan amount, first repaid date, providing insights into repayment behavior and creditworthiness.
3. Performance Data: Information on specific loan applications, including loan amounts and a target variable indicating loan performance (good/bad flag).



1 Methodology

1.1 Data Processing

Data cleaning was done to handle missing values, outliers and remove duplicates. Exploratory Data Analysis (EDA) enabled us understand the features and their relationships. New features were derived to improve model performance. Feature scaling was not adopted since the models used were not impacted by data scales.

1.2 Data Processing

Exploratory Data Analysis (EDA): Conducted EDA to understand data relationships, involving data profiling with YData profiling and the basic exploration of data using python EDA methods and functions such as info(),shape(), describe etc. Figure of data imbalanced at a ratio of 1:4. smote techniques was used to selected data for model training

Data Cleaning: Removal of duplicates and imputation of null values was done.

Feature Engineering: New features were derived from the provided columns and merged with the other data frames to enhance model performance. The columns created include the following;

1. age category
2. education flag
3. employment flag
4. bank account type flag
5. loan frequency
6. average time taken to pay back
7. loan clients avg loanAMT, clients max loan AMT
8. referred client
9. ontime payment percentage
10. pays late risk
11. ontime loans
12. percentage loan interest
13. Borrower Reliability Score

1.3 Model Selection and Training

1. Algorithm Exploration: A study was undertaken to identify, the best suited algorithm for the data provided, and the nature of problem at hand. The team zeroed on classification algorithms; Random Forest, LightGBM, XGBoost, and Neural Networks and these were implemtened.
2. Hyper-parameter Tuning and Training: This was done across the various models, the prediction model was retrained and tested. XGBoost gave the best performance results and was therefore adopted.

3. Data Splitting: The dataset was pre-divided into training and validation sets to prevent over fitting and model contamination.
4. Model Evaluation: Models were evaluated on the validation set using metrics such as accuracy, F1 score, precision, recall, and AUC.

Feature Importance: Analysis was conducted to determine the most influential features.

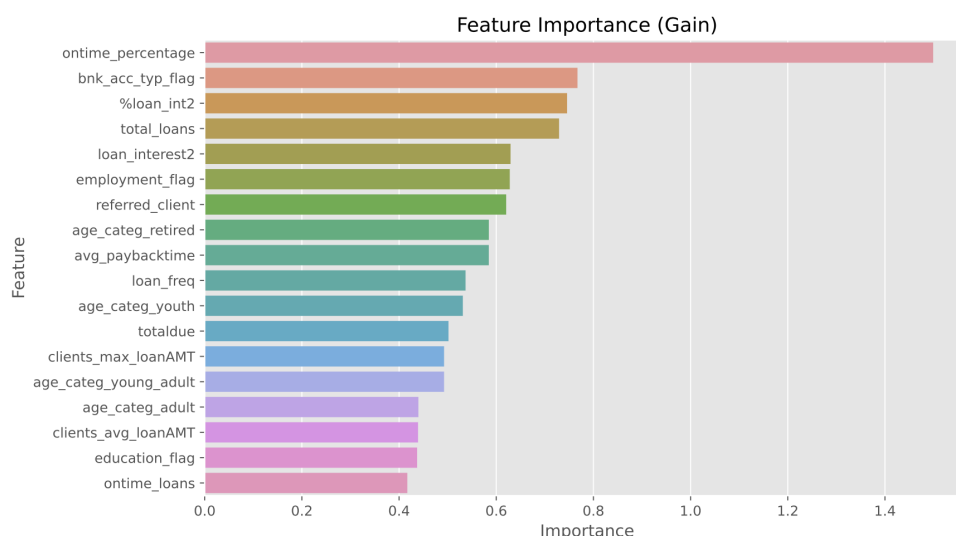


Figure 1: Feature importance

From the Feature importance figure, the top features included percentage of loans paid ontime, interest rate, bank account type.

2 Results

2.1 Model Evaluation

Performance Metrics: All models were assessed using accuracy, F1, recall, precision, and AUC. XGBoost achieved the best accuracy score of 0.77, leading to its adoption.

```
accuracy score: 0.7368421052631579
f1 score: 0.8389355742296917
precision score: 0.7881578947368421
recall score: 0.8967065868263473
auc: 0.6122064124178827
```

```
Perf result Xtrain using smote data – poor pe
accuracy score: 0.7093821510297483
f1 score: 0.812407680945347
precision score: 0.8017492711370262
recall score: 0.8233532934131736
auc: 0.621798877972211
```

```
Perf result Xtrain excludes features of 0.1 a
accuracy score: 0.7356979405034325
f1 score: 0.8381219341275403
precision score: 0.7878787878787878
recall score: 0.8952095808383234
auc: 0.6135907796058369
```

```
Perf result Xtrain.....
accuracy score: 0.7700228832951945
f1 score: 0.8659106070713809
precision score: 0.7809867629362214
recall score: 0.9715568862275449
auc: 0.6135907796058369
```

Figure 2: A Screenshot of performance results

- Accuracy Score: 0.7654462242562929 This is the ratio of correctly predicted instances to the total instances. An accuracy of 0.765 implies that approximately 76.5 percent of the predictions made by the model are correct.
- F1 Score: 0.8634243837441706 The F1 score is the harmonic mean of precision and recall. It provides a balance between the precision and the recall, and it is especially useful when you need to take both metrics into account. An F1 score of approximately 0.863 suggests that the model has a good balance between precision and recall.
- Precision Score: 0.7779111644657863 Precision is the ratio of correctly predicted positive observations to the total predicted positives. A precision of about 0.778 indicates that roughly 77.8 percent of the instances predicted as positive are actually positive.
- Recall Score: 0.9700598802395209 Recall (or sensitivity) is the ratio of correctly predicted positive observations to the all observations in the actual class. A recall of about 0.970 suggests that the model correctly identifies 97 percent of the actual positive instances.
- AUC (Area Under the Curve): 0.6122064124178827

AUC represents the degree or measure of separability between the classes. It tells how much the model is capable of distinguishing between positive and negative classes. An AUC of 0.612 suggests that the model has a relatively low ability to distinguish between the positive and negative classes. Below is a picture of results of various model implementations. SMOTE technique was applied to manage the data imbalance, however the model performance didn't improve and therefore the complete train dataset was used.

2.2 Model Deployment and Refinement

Final Model Selection: The group chose the best-performing model from the validation stage. Deployment: The prediction results from the model were submitted to Zindi platform and will also be deployed on hugging face.

2.3 Monitoring and Improvement

The team will continue to improve the model so as to improve its accuracy.

2.4 Leader board Performance

The model prediction was scored at 0.215 by Zindi, ranking 123rd on the competition leader board.

Competition Leaderboard

Unless stated otherwise in the Info Page, this leaderboard reflects scores based on only a portion of the total test set until the competition closes. See competition Info for more information.

RANK	USER	PUBLIC SCORE
123	 Eulers Team	0.215862068

Figure 3: A Screenshot of ZINDI Leader board

3 Discussion

The Euler Group's project successfully addressed the critical issue of loan fraud detection through the development of a predictive model using advanced machine learning techniques. The project involved comprehensive data processing, rigorous feature engineering, and meticulous model evaluation, resulting in a model that significantly enhances the efficiency and reliability of loan processing systems.

3.1 Key Insights

1. Feature Engineering: The creation of features like Borrower Reliability Score played a crucial role in improving model performance.
2. Dropping features rated less important with least score of 0 and 0.1 also improved performance
3. Algorithm Selection: XGBoost's superior performance highlights the importance of choosing the right algorithm and tuning its parameters for optimal results. introduction of the Grid search impacted its performance
4. Evaluation Metrics: A holistic evaluation using multiple metrics ensured a robust assessment of model performance, addressing various aspects of accuracy and reliability.

3.2 Future Work

To further improve the system, future efforts could focus on:

- Incorporating additional data sources to enrich the dataset.
- Exploring more sophisticated algorithms and ensemble methods, deep learning algorithms.
- Enhancing the interpret ability of the model to better understand the decision-making process.